

MMI 805 — FOUNDATIONAL DATA SKILLS

Module 5: Visualising Multidimensional Data

MSc in Mathematical Innovation • Open University of Kenya

1 MODULE OVERVIEW

When a dataset contains three or more variables and we want to see how they relate simultaneously, standard two-dimensional plots become insufficient. This module introduces practical techniques for visualising multidimensional data in a single view, using colour, shape, size, faceting, and layered geometric objects (geoms). We also examine why statistical software — particularly `ggplot2` — is dramatically more capable than spreadsheets for producing these complex visuals clearly and reproducibly.

LEARNING OUTCOMES — By the end of this module, you will be able to:

1. Describe the main techniques for visualising data with three or more dimensions (aesthetics and facets).
2. Explain the advantages of statistical software over spreadsheets for creating complex, multidimensional visuals.
3. Create layered plots that combine multiple aesthetic mappings and faceting to communicate multivariate patterns.
4. Evaluate whether a multidimensional visualisation communicates its insights effectively or obscures them.

2 BEYOND TWO DIMENSIONS

2.1 1.1 The Dimensionality Problem

A standard scatter plot represents two variables: one on the x -axis and one on the y -axis. But most real analytical questions involve more than two variables. How does the relationship between child age and height-for-age vary by sex? Is the effect of household wealth on vaccination coverage different in rural vs. urban areas? How do three or more variables jointly explain an outcome?

Statistical graphics solve this problem by mapping additional variables to visual **aesthetics** beyond x and y position: colour, shape, size, transparency, and line type. Additionally, **faceting** — creating separate small panels for each level of a variable — allows an entire relationship to be replicated across subgroups.

2.2 1.2 Aesthetic Mappings in `ggplot2`

In `ggplot2`, aesthetic mappings connect data variables to visual properties:

x, y	Any	Position on axes; the primary mappings.
colour	Categorical or continuous	Hue for categories; colour gradient for continuous. Use colour-blind-safe palettes.
fill	Categorical or continuous	Interior fill for bars, boxes, areas.
shape	Categorical (≤ 6 levels)	Different point shapes per category. Avoid for continuous variables.
size	Continuous	Point or line width; effective for a third continuous variable.
alpha	Continuous	Transparency; useful for overplotting in dense scatter plots.
linetype	Categorical	Solid, dashed, dotted lines for different categories.

2.3 1.3 Faceting

Faceting creates a grid of panels, each showing a subset of the data defined by one or more categorical variables. This is often more readable than mapping a third variable to colour when there are many categories.

Listing 1: Multidimensional visualisation with ggplot2

```

1 library(ggplot2)
2 library(dplyr)
3
4 # EXAMPLE 1: Three variables using colour and size
5 # x = household wealth score, y = HAZ, colour = mother's education
6 # size = number of children in household
7 ggplot(mics_u5, aes(x = hh_wealth, y = haz,
8                     colour = mother_educ, size = n_children)) +
9   geom_point(alpha = 0.4) +
10  scale_colour_viridis_d(name = "Mother's Education") +
11  labs(title = "Wealth, Maternal Education, and Child Nutritional
12         Status",
13        x = "Household Wealth Score", y = "HAZ Score",
14        size = "No. of Children",
15        caption = "Source: MICS Kenya 2014")
16
17 # EXAMPLE 2: Faceted plot - relationship by region and urban/rural
18 ggplot(mics_u5, aes(x = child_age_months, y = haz)) +
19   geom_point(alpha = 0.2, colour = "#037b90") +
20   geom_smooth(method = "loess", colour = "#ff7f50") +
21   facet_grid(region ~ urban_rural) +
22   labs(title = "Age-HAZ Relationship by Region and Residence Type")

```

```

22     x = "Child Age (months)", y = "HAZ Score")
23
24 # EXAMPLE 3: Layered geoms (points + boxplot + jitter)
25 ggplot(mics_u5, aes(x = mother_educ, y = haz, fill = mother_educ))
26   +
27   geom_violin(alpha = 0.4) +
28   geom_jitter(width = 0.1, alpha = 0.1, colour = "#2B7A78") +
29   stat_summary(fun = median, geom = "point", size = 3, colour = "#
    ff7f50") +
30   coord_flip() +
31   labs(title = "Distribution of HAZ by Mother's Education Level")

```

3 SPREADSHEETS VS. STATISTICAL SOFTWARE FOR COMPLEX VISUALISATION

3.1 2.1 Why Spreadsheets Struggle with Multidimensional Visuals

Excel can produce a chart with colour mapped to a category, but this requires manual steps that are:

- **Not reproducible:** The same chart cannot be recreated automatically from the data.
- **Fragile:** Changing the data requires manually recreating the chart.
- **Limited:** Excel cannot natively produce faceted plots, violin plots, or properly layered geoms.
- **Error-prone:** Colour assignments can shift if data is sorted differently.

In R, a multidimensional chart is produced by a single, 10-line code block that automatically updates if the data changes, is reproducible by anyone who runs the code, and handles thousands of data points gracefully.

3.2 2.2 Evaluating Multidimensional Visual Effectiveness

A multidimensional visualisation is effective if:

- The reader can immediately identify what each visual channel (colour, size, facet) represents.
- The insight claimed in the caption is clearly visible in the chart without requiring mental calculation.
- No aesthetic is used for a variable that would be better represented by a different channel.
- The chart does not overload the reader with too many simultaneous mappings (practical limit: 3–4 variables per chart).

4 LEARNING ACTIVITIES

4.1 3.1 Video and Reading

- Watch: Introduction to multidimensional visualisation in ggplot2 (course platform).
- Complete the two STACK quizzes on aesthetic mappings and faceting.
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*. Free: <https://clauswilke.com/dataviz/>

4.2 3.2 Forum Activity

Post to the discussion forum: one layered plot and one faceted plot produced from the MICS Kenya data. Each plot must encode at least three variables. Include a caption and a one-paragraph explanation of what the visualisation reveals. Respond constructively to at least two classmates.

4.3 3.3 Module 5 Assignment

MODULE 5 ASSIGNMENT: MULTIDIMENSIONAL VISUAL BRIEF

Using the MICS Kenya dataset, produce a **visual brief** (PDF, maximum 600 words plus figures) containing:

1. **Three figures**, each encoding at least three variables using a combination of aesthetics (colour, size, shape, or alpha) and/or faceting.
2. For each figure: a descriptive title, axis labels with units, data source, and a 3–5 sentence caption explaining what the figure shows.
3. **Design notes** (100–150 words total): For each figure, one sentence explaining your aesthetic choices and why you preferred this mapping over alternatives.
4. **Insights** (150–200 words): What do your three figures collectively tell you about relationships in the data?
5. **Reproducibility note**: Include your R code (as an appendix or in-line).

At least one figure must use faceting. At least one must use a continuous colour or size mapping.

5 KEY CONCEPTS AND TERMINOLOGY

Aesthetic Mapping	A connection between a data variable and a visual property (position, colour, shape, size, alpha) in a <code>ggplot2</code> plot.
Faceting	Creating a grid of small panels, each showing a subset of data defined by one or more categorical variables.
Geom	A geometric object used to represent data in <code>ggplot2</code> (e.g., <code>geom_point</code> , <code>geom_boxplot</code> , <code>geom_violin</code>).
Layered Plot	A <code>ggplot2</code> plot combining multiple geometric objects to show the same data from different perspectives simultaneously.
Overplotting	A visualisation problem where too many data points overlap, obscuring the density of the distribution. Addressed with alpha transparency or jittering.
Grammar of Graphics	A framework (Wilkinson, 2005; Wickham, 2010) for constructing statistical graphics by specifying data, aesthetic mappings, geometric objects, and scales.

MODULE 5 SUMMARY

- Multidimensional visualisation maps additional variables to visual channels beyond x/y position: colour, shape, size, transparency, and faceting.
- Statistical software (`ggplot2`) dramatically outperforms spreadsheets for multidimensional visuals: reproducible, flexible, extensible, and scalable to large datasets.
- Effective multidimensional charts are legible (3–4 variables maximum), clearly labelled, and show patterns not visible in simpler views.
- Faceting is often more readable than colour for categorical variables with more than 4–5 levels.
- *A good visualisation does not just display data — it reveals a story that is difficult to see in numbers alone.*